The Study of Baby Crying Analysis Using MFCC and LFCC in Different Classification Methods

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***Abstract*—** Nowadays, there are so many researches about the baby crying detection for many purposes. The right interpretation of the crying baby is notable for the medical objective so caregiver knows how to treat baby well. Babies within the first three months of age use Dustan Baby Language (DBL) to communicate. Based on some researches, there are five words to express their needs, such as "Neh" (I am hungry), "Eh" (burp is needed), "Owh/Oah" (fatigue), "Eair/Eargghh" (cramps), "Heh" (physical discomfort; feel hot or wet). Beside that purpose, smart home technology implements baby crying detection for monitoring the baby. Several stages to detect crying baby are preprocessing, feature extraction, and classification. Popular feature extractions for voice or sound recognition are Mel Frequency Cepstral Coefficient (MFCC) and Linear Frequency Cepstral Coefficient (LFCC). In this study, both of that feature extraction have been analyzed to know the appropriate condition when using one of those feature extractions. Classification methods (KNN classification, Vector Quantization, and Simple Neural Network) affect the accuracy in detecting and recognizing of baby crying. KNN classification with LFCC results better accuracy than using MFCC one with the sample data is female voice. If using baby voice, there is no significant different accuracy in both of feature extractions.

Keywords— DBL, MFCC, LFCC, KNN, VQ.

I. INTRODUCTION

Research on the baby crying consist of detecting the sound of the crying baby and recognizing or identifying the baby's need when he cries (baby crying translation). Baby crying detection is implemented on the smart home technology to monitor the baby easily. Parents no need to always inspect the CCTV when the baby's home with the babysitter, but they can get notification when their baby's crying automatically. For a medical objective, it is important to know what baby needs from his crying. Before crying, the baby will try to communicate with specific language which known as Dustan Baby Language (DBL) that has some meaning like "I am hungry", "I am sleepy", and others. The baby language is grouped into five meanings that use as universal language of babies. The sound of crying baby contains a lot of information about his emotional and physical condition, and also the baby identity. Priscilla Dustan found that infants with the first three months of age using proto language to communicate, which is five words to express their needs [1]. That five words are "Neh" (hungry), "Eh" (need to burp), "Owh/Oah" (fatigue), "Eair/Eargghh" (cramps), "Heh" (physical discomfort; feel hot or wet). Fundamental frequency in baby crying is ranging from 250 Hz to 600 Hz [2].

Crying baby detection by Dunstan Baby Language (DBL) is through three main stages, the first is preprocessing to normalized all sound data, the second is feature extraction, and the last one is classification process. The popular feature extraction methods for audio processing are Mel Frequency Cepstral Coefficient (MFCC) and Linear Frequency Cepstral Coefficient (LFCC), they use sound frequency based. For the classification process, there are so many methods that have been used, but in this study, KNN classification, Vector Quantization (VQ) and Simple Neural Network (SNN) are analyzed to inform the appropriate condition when using this method combination from feature extraction to the classification.

II. BASIC THEORY

A. Speech Signal

The process of forming speech signals starting from the larynx (where the vocal cords located) and ended up in the mouth. Speech or voice signals categorized into voiced and unvoiced. Unvoiced is a condition where the state of the vocal cords do not vibrate. Voiced is a condition where the state of the vocal cords vibrate and produce a pulse of the glottis. Pitch is known as the fundamental frequency of the glottis [3]. The human voice has a low-frequency range with a fundamental frequency about 220 Hz for women and 130 Hz for men and for the first formant vocal discrimination under 1000 Hz [4].

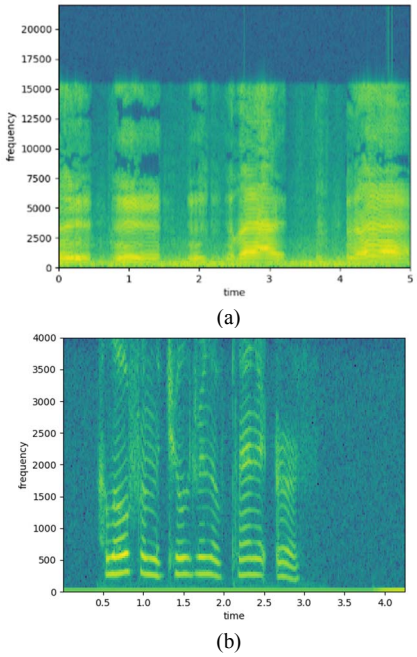


Fig. 1. (a) Spectogram of baby crying sigmal; (b) Spectogram of speech sound signal

Adult voice and the sound of crying babies have similarities and differences, in previous studies were found

differences in the character of the sound in the aspect of the fundamental frequency (pitch) where the voice of crying babies is higher. The voice of crying babies has short vocal cords and thin so it can be seen from spectrogram have tidy characters[5]. A speech signal classification system should be able to categorize different types of input voice, especially to detect the type of speech, noise or music genre[3].

B. Dunstan Baby Language (DBL)

Priscilla Dunstan proposed the idea to identify the meaning of a crying baby called Dunstan Baby Language [2], That there are five types of universal sound of crying babies and their meanings are as follows:

"Neh": The sound of "Neh" comes from sucking and tongue pushed into her mouth, which means that the baby is hungry.

"Owh/Oah": Sound "Owh" would sound like a man who yawns which means that the baby was sleeping.

"Heh": The sound of "Heh" is derived from the infant's response from burning or itching, which means that the baby was not comfortable.

"Eairh/Eargghh": Sound "Eairh" is generated when the baby does not burp which causes air bubbles enter the stomach and can't be released, this means that the baby is experiencing gastric problems.

"Eh": Sound"Eh" is generated when the wind get trapped and does not get out in the chest that causes the air bubbles out of the mouth, this means the baby wants to burp.

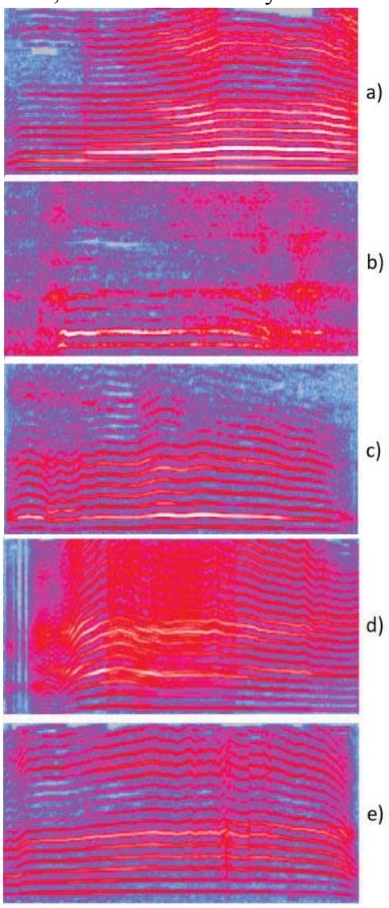


Fig. 2. Spectogram of baby crying signal : a) Neh; b) Heh; c) Eh; d) Eairh/Eargghh; e) Owh/Oah [1]

C. Feature Extraction Method

After pre-processing, the next step of speech-recognition systems is feature extraction. It is an important process to get the feature of the audio which can distinguish one from the other. Audio's features are extracted by dividing the input signal from the frame with a length of 10-40ms then each feature value is calculated [3]. Many studies of speech recognition system use MFCC feature extraction because it is considered similar to the concept of human hearing [6]. Other studies also use LFCC because it has similarities with the concept of MFCC.

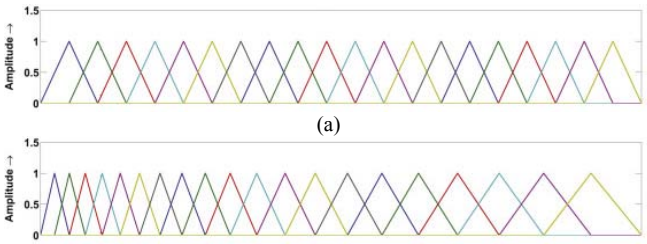


Fig. 3. (a) Linear filter-bank; (b) Mel filter-bank [7]

Mel-Frequency Cepstral Coefficient (MFCC)

MFCC is a method of extracting feature that converts sound into a voice signal vector. This method provides a representation of the short-term power spectrum of the signal. MFCC concept was similar to human hearing which has a critical bandwidth of the human ear at the frequency below 1000Hz. MFCC process start from dividing the sound signal into the form of a frame with a duration of 10-40 milliseconds time frame, this is the frame blocking. Then frame blocking is windowed by hamming to eliminate aliasing effects that occur due to the framing. The windowing process where w_n is a windowing function and N is a number of samples in one frame, the equation (1) is the formula for the windowing process:

$$w_n = 0.54 - 0.46\ \cos\frac{2\pi n}{N-1},\quad 0 \le n \le N-1 \qquad (1)$$

The results of this windowing process then followed by the calculation of the Fast Fourier Transformation (FFT) which converts the signal from the time domain into the frequency domain. Filter-bank applied to the signal with the frequency domain so that signal it turns into Mel frequency by the equation (2):

$$\mathrm{Mel(f)} = 2595\ \log_{10}\left(1 + \frac{f}{700}\right) \qquad (2)$$

MFCC using bank filter Mel-scale which is a band-pass filter triangle logarithmic [8]. It makes a higher frequency filter results the greater bandwidth. The final stage in the MFCC process is Discrete Cosine Transformation (DCT) after the results of the previous process is converted back into the time domain, so that the signal can be presented well. These results form a row of an acoustic vector which named Mel Frequency Cepstral Coefficient.

Based on previous literature, the human ear has non-linear characteristics in tone perception. The relationship between Frequency and Mel-scale described for a case about frequency above 500Hz where the increasing value of the interval is proportional to the increasing in the same pitch, for example, four octaves on a scale Hz over two octaves on the Mel scale value [3]. The relationship between the use of Mel scale and

scale Hz in a nonlinear mapping function is useful for analyzing seismic signals where there are few differences between the speech signal and seismic signal. The filter is used for the frequency range 0-22050 Hz in speech recognition but in the case study took samples on the band seismic signal is below 500Hz. Mapping function which gets under 1000Hz relatively linear so that the use of MFCC work is not good enough at frequencies below 1000 Hz [9].

Linear Frequency Cepstral Coefficients (LFCC)

LFCC is a feature extraction method which laid out well. The process starts with breaking the audio clip into multiple segments consisting of a fixed number of frames. LFCC has extraction process characteristic that similar to MFCC [10]. LFCC using a linear filter-bank to replace the Mel filter-bank. The use of linear filter-banks is very well used in the high-frequency area.

D. Classification Method

After getting the feature value from the feature extraction process, these values will be calculated by the classification method used.

Vector Quantization

VQ is a method to map a large number of vectors of the space to the number of clusters defined by each center is represented as a vector. VQ produce lower distortion. Where most of the value of the feature vector embraced and a set of vectors by the small size produces a value that matches to the centroid of distribution [2].

K-Nearest Neighbor Algorithm (KNN)

KNN is a method that focuses on the classification of types of data to other data that has its own label vector. This classification determines the non-linear decision boundary to increase its performance. Here is a distance metric that is often used to calculate the distance of the sample, the Euclidean distance. Using two samples x and y, Euclidean distance between the samples was determined by the formula (3):

$$|x - y| = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \quad (3)$$

where n is the number of features that describe the x and y [3].

III. SYSTEM DESIGN & OVERVIEW

Baby crying detection system focuses on monitoring the baby. The system can be implemented in a smart home so that caregivers or parents can monitor their children. This system can be explained in figure (4):



Fig. 4. System overview

Several stages in the system are as follows:

1. Voice Input : Voice input consists of various kinds of children's voice signals which are digital evidence that is tested whether it is a baby's cry or not. Before entering the extraction stage, the sound is pre-processed to remove signals from the noise or other unexpected sounds.
2. Feature Extraction : Feature Extraction is a stage that aims to convert voice signals using digital signal processing so that the signal can be differentiated by the system. At this stage, MFCC and LFCC are used as feature extraction methods.

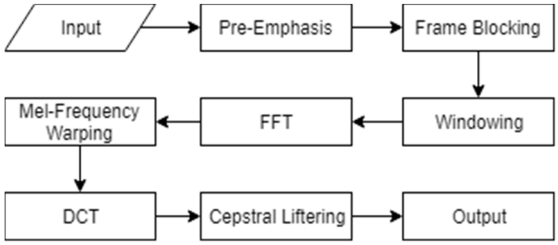


Fig. 5. MFCC Feature Extraction

The stage of MFCC can be seen in figure 5, LFCC has similar steps to MFCC, the different only in the type of fillter-bank used.

3. Classification : Classification at this stage is a continuation of the feature extraction stages. The sound signal will have its own characteristics so that at this stage the sound signal can be classified.
4. Data analysis and matching process are conducted after the sound signals are classified. At this stage, all samples and classification results are analyzed.

IV. RESULT AND DISCUSSCION

The previous study, MFCC is good enough in speech recognition, but it is bad enough when the audio contains a lot of noises [10], so it needs proper preprocessing to eliminate the noise. In an experiment to classify the samples of nine-speaker using MFCC and VQ codebook, the process to get cepstral value is calculated using 12 coefficients of the nine different sounds. The database consists of 21 forms a sound signal, 8 of them from different users and the rest from the same users. There are seven women's voices and men. The test is carried out at a noisy place results a high failure rate of MFCC, i.e. 20% where the failure occurred in testing the sound Speakers8-Male detected as Spekers-9-Female. Then using LFCC as the addition of MFCC method can reduce the error rate [11]. Table I shows the change in the value of the error before and after adding LFCC method in feature extraction MFCC from 146 speakers (73 male and 73 female). First row is EER with 10 common sentences, and the second row is unique sentences testing.

TABLE I. EER (EQUAL ERROR RATIO) BEFORE AND AFTER LFCC

BEFORE		AFTER	
MALE	FEMALE	MALE	FEMALE
2.63 %	7.01 %	2.28 %	3.48 %
2.49%	5.9 %	1.96 %	3.22 %

In one test to the case of a telephone conversation it was found that LFCC gave EER increase 21.5% and 15.0% relative to MFCC in nasal and non-nasal consonants area [6]. Performance comparison between MFCC and LFCC using NIST SRE 2010 shows that the MFCC and LFCC complete

each other. LFCC is better than MFCC in capturing spectral in high frequency regions, such as in detecting female voice as long as other parameters are same. It is because the female's vocal channel is relatively shorter and its formant frequency is higher than male's one [12].

A. Analyze using MFCC

MFCC has some advantages in extracting feature that used for analysis of baby crying classification [13], such as :

- It can identify the character of the sound so that it can determine the pattern of sound.
- The output vector has a small data size but does not remove the noise characteristics in the extraction.
- MFCC works similar to the way of a human listener works in giving their perceptions.

The test using KNN classification method with the value K=1 when compared to the Simple Neural Network with two hidden layers, the first layer has seven knots. The accuracy of recognition shows in table II.

TABLE II. ACCURACY KNN VS SNN WITH MFCC

Type of cry	Classification Method	
	KNN	NN
Neh	80.00%	40.00%
Owh	100.00%	80.00%
Heh	66.67%	100.00%
Eairh / Eh	57.14%	42.68%
Average Accuracy	75.95%	65.67%

Table II is the result of MFCC feature extraction using tuner package has several parameters for the input data. The training data is 85% of 4 classes (Eairh / Eh, Heh, Neh, and Owh) of baby crying; using KNN K = 1; and normalized standard deviation for the number of sample testing is 139 audio[14]. Sample Rate recording = 11205 Hz; Total spectra (n bands) = 40; Window time = 0.02 to 0.08 seconds; The distance between frames (hop time) where if the voice recording is less than 2 seconds than set the value is 0.1 seconds hop time reduce the duration of the recording, but when recording more than 2 seconds, the value is 1 second hop time reduced the duration of the recording; Pre-emphasize = 0.97; Sum power = true; Liftering exponential value (lifterexp) = 0.6; Spectral bandwidth (bwidth) = 10; Usecmp = True; Number of columns in the result vector (numcep) = 10 columns for each hop time where the number of features produced in the extraction process is 20 Frequency Features. The average accuracy of 96.67% is found in the system if the parameter Wintime = 0:08;

Unlike the case with the use of five types of cries by presenting 50 data samples consist of 40 training data samples and 10 testing data samples for each class. The accuracy of all recognition: Neh is 60%, Eh is 70%, Owh is 60%, Eairh is 70%, and Heh is 70%. Where the signal was recorded for 20 sec at the hospital environment [8]. The overall average accuracy is 66%.

MFCC has high accuracy using Euclidean distance parameter. This model can produce accuracy about 94% [15]. This happens because the signal cut research sound of silence at the beginning and end of the speech signal. The conclusion is that using MFCC method has high accuracy values when using KNN.

B. Analyze using LFCC

Using codebook model, LFCC and Euclidean distance, where the voice signal is extracted using parameters of a pitch but no recorded using a Vector Quantization, it yields accuracy about 93% to detect crying baby[2], The benefits of using this method is :

- It is easy to identify babies who cry and to verify the use of KNN to classify infant emotions.
- It results high accuracy if using euclidean distance.
- It provides higher accuracy in detecting emotion from a crying baby.
- Cutting silent voice signals to produce a sound that is more specific resulting higher accuracy.
- LFCC produces the same frequency with the MFCC[16].

C. Comparison of MFCC and LFCC

There are 40 data that are baby crying and 40 data are not baby crying consisting of noise, mute sounds and baby laughter as training data. then the data were tested for each of the 10 data in each voice category (Crying and Non-Crying) using the LFCC and MFCC methods using the KNN classification with K = 3, the results are shown in Table III.

TABLE III. ACCURACY LFCC VS MFCC WITH KNN

Test Case	LFCC	MFCC
Crying	90%	80%
Non-Crying	90%	90%
Average Accuracy	90%	85%

Tests were conducted to the baby voice between the ages of 0-9 months [17]. There are noise additions such as engine noise, passers-by, and motorcycles. There are 140 training data with 5 categories with 28 samples each. Number of testing data is 35 consist of 7 crying babies for every kind of crying. Testing was conducted to analyze performance MFCC and LFCC. There is significant performance between both methods, the average percentage of LFCC accuracy is about 91.58% and the average percentage of MFCC accuracy is about 82.14% [15]. Recognition rate (number of recognizable words / number of words presented) of each class of crying baby sound is shown in table IV [18].

TABLE IV. THE RECOGNITION RATE LFCC VS MFCC

Test Case	LFCC	MFCC
Eairh/Eh	87.22%	85.11%
Heh	88.19%	86.87%
Neh	94.57%	79.89%
Owh	93.33%	80.87%
Average Accuracy	90.83%	83.19%

It can be concluded based on Table III and Table IV that percentage of using LFCC is better than MFCC although both methods have accuracy values above 80% [17], It is because the sound of crying babies is on high-frequency region.

The use of LFCC combined with the VQ codebook using BG calculation with the data collected from the sound of babies aged 0-6 months. 150 voice as training data that represent each of the 50 cries each category and testing data as many as 40. LFCC works well in capturing high-frequency area so it concluded LFCC better than MFCC [18], However, both these methods are considered either to be applied in analyzing the emotions in the baby's crying.

The equation below will produce the Mel scale cepstral coefficient. Where the value N is the number of Mel Frequency Wrapping, L is the number of Mel-Scale Cepstral Coefficient.

$$C_m = \sum_{k=1}^{N} \cos\left[m\left(k-\frac{1}{2}\right)\frac{\pi}{N}\right] E_k, m = 1,2,\ldots,L \quad (4)$$

The experiment of baby crying sound uses L or number of cepstral = 19 results cepstral figure of MFCC and LFCC as shown in figure 6 and 7 with 19 different colors who represent the 19 cepstral.

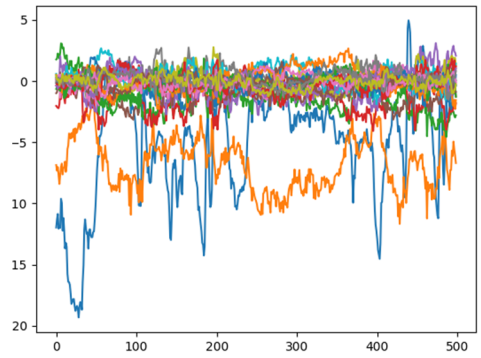


Fig. 6. MFCC with 19-Cepstral

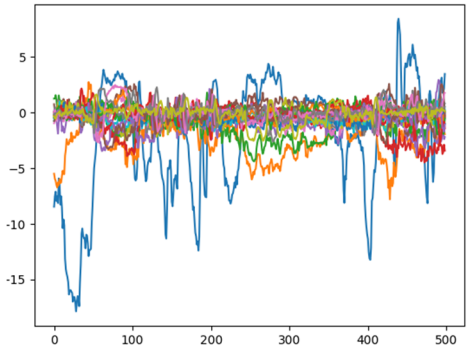


Fig. 7. LFCC with 19-Cepstral

Figure 8 is a combination from figure 6 and 7 that represent MFCC with orange line and represent LFCC with blue line. MFCC produces a less stable pattern compared to LFCC pattern. Where the sound of baby cries on high-frequency area on the LFCC is better than MFCC as shown on figure below. But, it is not significant different accuracy on baby voice than adult voice because the characteristic of baby voice not as much as adult voice.

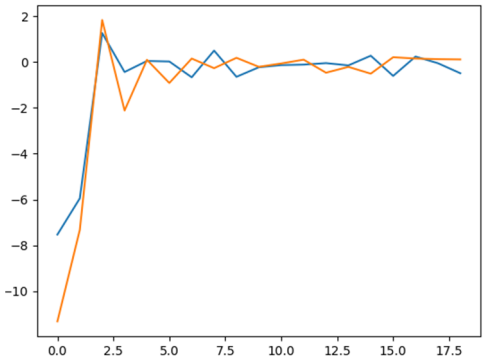


Fig. 8. Final result of LFCC and MFCC

V. CONCLUSION

Automatic crying detection methods for infants with high fundamental frequency (pitch), with short and thin vocal cords in crying baby sound using LFCC feature extraction and k-Nearest Neighbor (KNN) algorithm for classification is more effective than using MFCC and 2 other classifications (SNN and VQ). LFCC uses the linear cepstral coefficient, while MFCC uses a filter-bank which is a logarithmic triangle band-pass filter. Because of its filter bank characteristic, the use of MFCC is relatively not good enough high frequency of voice such as female voice and baby voice so it is recommended use LFCC. LFCC outperforms MFCC when using female voice trial data. It is because the female vocal tract is relatively short and the formant frequency obtained is relatively high. Besides that, the use of LFCC feature extraction as an addition to the MFCC method can help reduce the error rate at MFCC.

The accuracy value can be higher because of the preprocessing factor where the mute sound signal is cut at the beginning, unvoiced sound, and end of the sound, so that the feature more valuable and precise. The choice of the use MFCC method in a condition that full of noise is considered to be unsuitable, but it is still good in performance if proper preprocessing conducted and the voice is in regular frequency. The accuracy of the value on the use of the LFCC and MFCC methods depends on the number of test samples used and the type of sample being tested.

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